

AP CSP | Lesson 11: Selection

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Trace statements in their execution order. Each program statement and each brace occupies its own line. Show reasoning and use the stated valid input range.

1. When minutes is 30, which pair gives the values of (minutes $\geq$ 30, minutes $>$ 30)? [1 mark]

A. (true, true) B. (true, false) C. (false, true) D. (false, false)

Answer: _____

2. A program reaches an IF whose condition is false. There is no ELSE block. What happens next? [1 mark]

A. The program stops. B. The IF block executes once.
C. The IF block is skipped and execution continues after it. D. The program waits until the condition becomes true.

Answer: _____

3. Initially x is 4. What does this code display? [1 mark]

```
IF(x  $\geq$  4)
{
  x  $\leftarrow$  x + 3
}
DISPLAY(x)
```

A. 4 B. 3 C. 11 D. 7

Answer: _____

4. An IF condition is true. Its block changes the tested variable so that the same expression would now be false. What happens to this IF statement's ELSE block? [1 mark]

A. It remains skipped. B. It runs immediately after the first block.
C. Both blocks restart. D. It runs only if DISPLAY is present.

Answer: _____

5. Which TWO statements are correct? Select exactly TWO. [1 mark]

A. Two separate IF statements always behave like IF / ELSE. B. A separate later IF uses the current values when it is reached.
C. IF / ELSE chooses one of its two blocks from a single condition evaluation. D. A false condition automatically resets its variable to zero.

Answer: _____

6. A score starts at 12. The first IF tests score $\geq$ 10 and sets score to 5 when true. A second, separate IF then tests score $<$ 10 and adds 2 when true. What is the final score? [1 mark]

A. 5 B. 12 C. 7 D. 14

Answer: _____

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Written responses | Questions 7-8

7.

[4 marks]

Consider the code.

```
x ← 6
```

```
IF(x > 4)
```

```
{
```

```
  x ← x - 3
```

```
}
```

```
DISPLAY(x)
```

(a) State the condition value and the displayed value.

(b) If the first line becomes $x \leftarrow 4$, state the displayed value and explain why DISPLAY still executes.

8.

[4 marks]

Write AP pseudocode that receives a valid integer number of minutes from 0 to 180. Display "Goal met" for at least 30 minutes; otherwise display "Keep going". Then give the expected outputs for inputs 29 and 30.

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Written responses | Questions 9-10

9.

[4 marks]

A value starts at 12. First, if it is at least 10, a program sets it to 5. Next, a separate IF adds 2 if the current value is below 10.

(a) Give the value after the first block and the final value.

(b) If the second IF is replaced by the ELSE of the first IF, what is the final value? Explain why the results differ.

10.

[4 marks]

A program should display "Pass" for scores at least 70 and "Retry" otherwise. Scores are valid integers from 0 to 100. A student mistakenly uses `score > 70`.

(a) Give a test that exposes this mistake, with the required and actual output.

(b) Write the corrected condition.

(c) Give one test strictly below 70 and its correct output.

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Written responses | Questions 11-12

11. [4 marks]

Consider the code.

```
minutes ← 35
```

```
met ← minutes >= 30
```

```
minutes ← 10
```

```
IF(met)
```

```
{
```

```
  DISPLAY("Goal met")
```

```
}
```

(a) State met immediately after its assignment and immediately before IF.

(b) State the output.

(c) Write one assignment immediately before IF that would update met for the current minutes.

12. [4 marks]

A shop receives a valid integer amount from 0 to 200. If the amount is at least 100, the discount is 20; otherwise it is 0. Write code to display amount minus discount. Give expected outputs for amounts 99 and 100.
